

Literature status: the weakly open Daugavet-thickness question

Run `fa_banach.001`; source arXiv:2005.02045

Packet Status

This packet records a literature-already-answered result, not an original construction from the run. The supporting paper explicitly identifies the same question from Haller–Langemets–Lima–Nadel–Rueda Zoca and gives a negative answer. The packet addresses only Question (b) in the Ivakhno-related list from the source paper.

Source Question

The source is Rainis Haller, Johann Langemets, Vegard Lima, Rihhard Nadel, and Abraham Rueda Zoca, *On Daugavet indices of thickness*, arXiv:2005.02045. In the final section, the authors ask:

$$\mathcal{T}(X) := \inf \{ r > 0 : \mathcal{C} \subseteq B(x, r) \text{ for some } x \in X \}.$$

A Banach space X is said to have the *r-big slice property* if every slice of B_X is of radius one [Iva06]. Observe that the *r-big slice property* of a Banach space X implies that $\mathcal{T}^s(X) \geq 1$. Moreover, if every slice of B_X has diameter two, then X has the *r-big slice property*. Ivakhno asked if the converse is true [Iva06, p. 96].

In view of Ivakhno’s question, given a Banach space X , the following questions make sense:

- (a) If $\mathcal{T}^s(X) \geq 1$, then does every slice of B_X have diameter two?
- (b) If $\mathcal{T}(X) \geq 1$, then does every nonempty relatively weakly open subset of B_X have diameter two?
- (c) If $\mathcal{T}^{cc}(X) \geq 1$, then does every convex combination of slices of B_X have diameter two?

A negative answer to (c) easily follows from our results, as the following remark shows.

Remark 3.6. Let $1 < p < \infty$ and Y be a Banach space with the Dau-

The part relevant here is Question (b):

$$\mathcal{T}(X) \geq 1 \implies ? \quad \text{every nonempty relatively weakly open subset of } B_X \text{ has diameter 2.}$$

Remark 3.9 of the same source paper says that Question (b) remains open after their JT_∞ -predual example.

Supporting Literature

The supporting paper is Abraham Rueda Zoca, *Diameter, radius and Daugavet index of thickness of slices in Banach spaces*, arXiv:2306.01467. Its introduction states that the question whether $\mathcal{T}(X) \geq 1$ implies that every nonempty relatively weakly open subset of B_X has diameter two remained open in [1, Remark 3.9], and that this is precisely one of the questions addressed in the note. Its final Remark 4.4 records the answer:

Proof of Theorem 1.2. The proof follows by the construction of this section and the arbitrariness of δ and $\varepsilon > 0$. ■

Remark 4.4. In [7, pp. 18, Question (b)] it is asked whether $\mathcal{T}(X) \geq 1$ implies that every non-empty relatively weakly open subset of B_X has diameter 2. Theorem 1.2 gives a negative answer in an extreme way: Given $\varepsilon > 0$ there exists a Banach space X with $\mathcal{T}(X) > 2 - \varepsilon$ and X still contains slices of diameter strictly smaller than 2.

Remark 4.5. In [7, pp. 18] it is pointed out that if X has the r-BSP then

Idea of the Proof

The source question asks whether the threshold $\mathcal{T}(X) \geq 1$ forces all relatively weakly open subsets of the unit ball to be diameter two. Rueda Zoca's later construction starts from a Banach space with the Daugavet property and changes the norm. The new norm keeps the convex-combination thickness index very large, so in particular $\mathcal{T}(X) > 1$, but it also creates a genuine slice of the new unit ball with diameter strictly smaller than two. Since every slice is relatively weakly open, this directly violates the conclusion asked in Question (b).

Formal Statement

Theorem. *Question (b) from arXiv:2005.02045 has a negative answer. More strongly, for every $\varepsilon > 0$ there is a Banach space X with $\mathcal{T}(X) > 2 - \varepsilon$ whose unit ball contains a nonempty relatively weakly open subset of diameter strictly smaller than 2.*

Proof. Rueda Zoca's Theorem 1.2 in arXiv:2306.01467 states that if Y has the Daugavet property, then for every $\varepsilon > 0$ there is an equivalent renorming $|\cdot|_\varepsilon$ such that the renormed space fails the r -big slice property, so its unit ball contains slices of diameter strictly smaller than 2, and

$$\mathcal{T}^{cc}((Y, |\cdot|_\varepsilon)) \geq 2 - \varepsilon$$

after choosing the auxiliary parameters in the construction sufficiently small. Remark 4.4 of the supporting paper restates the consequence in the exact language of the source question: for every $\varepsilon > 0$ there is a Banach space X with $\mathcal{T}(X) > 2 - \varepsilon$ and with slices of diameter strictly smaller than 2.

For completeness, here is the direct implication. The Daugavet thickness indices satisfy

$$0 \leq \mathcal{T}^{cc}(X) \leq \mathcal{T}(X) \leq \mathcal{T}^s(X) \leq 2$$

as recalled in the supporting paper. Thus a lower bound near 2 for $\mathcal{T}^{cc}(X)$ gives the same lower bound for $\mathcal{T}(X)$. Taking $\varepsilon < 1$, the renormed space has $\mathcal{T}(X) > 1$. But one of its slices has diameter strictly smaller than 2. A slice of the unit ball is a nonempty relatively weakly open subset of the unit ball. Therefore $\mathcal{T}(X) \geq 1$ does not force all nonempty relatively weakly open subsets of B_X to have diameter 2. □

Limitations

This packet covers only Question (b) from the three-part Ivakhno-related list in the source paper. Questions (a) and (c) were answered negatively inside arXiv:2005.02045 itself and are therefore same-paper extraction false positives for this run. The separate ℓ_1 -sum question on ℓ_1 -sum equalities for $\mathcal{T}$ and $\mathcal{T}^{cc}$ is not addressed here.

Literature And Duplicate Check

The run indexes were searched for arXiv:2005.02045, Daugavet indices of thickness, Daugavet index, weakly open subsets, Ivakhno, Rueda, and ℓ_1 -sum phrases. No existing packet or attempt covered this exact Question (b). The supporting paper arXiv:2306.01467 explicitly cites [1, Remark 3.9] and states that Theorem 1.2 gives the negative answer. Nearby Daugavet/ Δ -point papers arXiv:2307.10647 and arXiv:2301.04433 are adjacent but do not change the scope of this exact literature match.

No computational checks were needed; the packet is a source-to-source literature identification.

Human Review Recommendation

Review as an explicit literature-already-answered packet. The main checks are that the later paper's [7, Remark 3.9] is the Haller–Langemets–Lima–Nadel–Rueda Zoca question shown above, and that the small-diameter slice in the supporting theorem is indeed a relatively weakly open subset of the renormed unit ball.

References

- [1] R. Haller, J. Langemets, V. Lima, R. Nadel, and A. Rueda Zoca, *On Daugavet indices of thickness*, J. Funct. Anal. **280** (2021), article 108846; arXiv:2005.02045.
- [2] A. Rueda Zoca, *Diameter, radius and Daugavet index of thickness of slices in Banach spaces*, arXiv:2306.01467.